Section 2–2 Binary Numbers:

7. Convert each binary number to decimal:

(g) 1011010.1010

8. What is the highest decimal number that can be represented by each of the following numbers of binary digits (bits)?

(h) nine

9. How many bits are required to represent the following decimal numbers?

(g) 132

10. Generate the binary sequence for each decimal sequence:

(e) 64 through 75

Section 2–3 Decimal-to-Binary Conversion:

11. Convert each decimal number to binary by using the sum-of-weights method:

(g) 125

12. Convert each decimal fraction to binary using the sum-of-weights method:

(b) 0.246

13. Convert each decimal number to binary using repeated division by 2:

(f) 59

14. Convert each decimal fraction to binary using repeated multiplication by 2:

(b) 0.347

Section 2–4 Binary Arithmetic:

15. Add the binary numbers:

(e) 1001 + 101

16. Use direct subtraction on the following binary numbers:

(f) 11010 – 10111

17. Perform the following binary multiplications:

(d) 1001 × 110

18. Divide the binary numbers as indicated:

(b) 1001 ÷ 11

Section 2–8 Hexadecimal Numbers:

37. Convert each hexadecimal number to binary:

(d) $5C8_{16}$

40. Convert each decimal number to hexadecimal:

(f) 2890

42. Perform the following subtractions:

(b) $C8_{16} - 3A_{16}$

Section 2–9 Octal Numbers:

43. Convert each octal number to decimal:

(g) 163_8

46. Convert each binary number to octal:

(h) 10110000011

Bin → Dec

Sum of Weights

$\boxed{7}$ (g) $1011010.1010_2 \longrightarrow ?_{10}$

$= 2^6 + 2^4 + 2^3 + 2^1 + 2^{-1} + 2^{-3}$

$= 64 + 16 + 8 + 2 + 0.5 + 0.125 = 90.625_{10}$

$\boxed{8}$ (h) Highest possible decimal number that can be represented by n bits $\longrightarrow 2^n - 1$

$n = 9 \longrightarrow$ highest possible decimal number $= 2^9 - 1 = 511$

$\boxed{9}$ (g) # of bits needed to represent $132_{10} = ?$

With n-bits, we can represent decimal numbers in the range $0 : 2^n - 1$

So the question becomes: What is the value of n that guarantees that 132_{10} belongs to the range $0 : 2^n - 1$

$\Rightarrow 132_{10} \leq 2^n - 1$

$\Rightarrow 133 \leq 2^n$

$\Rightarrow \log_2 133 \leq n$

$\Rightarrow \dfrac{\log 133}{\log 2} \leq n \Rightarrow n \geq 7.055 \Rightarrow n = 8$

minimum value

10 (e) Generate the binary sequence for $64_{10} \rightarrow 75_{10}$

$64_{10} = 1000000_2$

$65_{10} = 1000001$

$= 1000010$

$= 1000011$

$= 1000100$

$= 1000101$

$= 1000110$

$= 1000111$

$= 1001000$

$= 1001001$

$= 1001010$

$75_{10} = 1001011_2$

---

11 (g) $125_{10} = ?_2$ using sum-of-weights

$= 64 + 32 + 16 + 8 + 4 + 1$

$= 2^6 + 2^5 + 2^4 + 2^3 + 2^2 + 0 + 2^0$

$= 1 \quad 1 \quad 1 \quad 1 \quad 1 \quad 0 \quad 1_2$

---

12 (b) $0.246_{10} = ?_2$ using sum-of-weights

$= 0 + 0 + 0.125 + 0.0625 + 0.03125 + 0.015625 + \ldots$

$2^{-3} \quad 2^{-4} \quad 2^{-5} \quad 2^{-6}$

$= 0.0011 \ldots$

13 (f) $59_{10} = ?_2$ using repeated division by 2

$59 \div 2 = 29$ 1

$29 \div 2 = 14$ 1

$14 \div 2 = 7$ 0

$7 \div 2 = 3$ 1

$3 \div 2 = 1$ 1

$1 \div 2 = 0$ 1

111011_2

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14 (b) $0.347 = ?_2$ using repeated multip. by 2

$0.347 * 2 = \underline{0}.694$

$0.694 * 2 = \underline{1}.388$

$0.388 * 2 = \underline{0}.776$

$0.776 * 2 = \underline{1}.552$

$0.552 * 2 = \underline{1}.104$

$0.104 * 2 = \underline{0}.208$

$0.208 * 2 = \underline{0}.416$

$0.416 * 2 = \underline{0}.832$

$0.832 * 2 = \underline{1}.664$

$0.0101100 01 \ldots_2$

15 (e)

$$\begin{array}{r} {\scriptstyle 1} \\ 1001 \\ +\ 0101 \\ \hline 1110 \end{array}$$

---

16 (f)

$$\begin{array}{r} {\scriptstyle 1\ 1\ 1} \\ 11010 \\ -\ 10111 \\ \hline 00011 \end{array}$$

---

17 (d)

$$\begin{array}{r} 1001 \\ \times\ 110 \\ \hline 0000 \\ 1001 \\ 1001 \\ \hline 110110 \end{array}$$

---

18 (b)

$$\begin{array}{r|l} & 001 \\ \hline 11 & 1001 \\ & 11 \\ \hline & 11 \\ & 11 \\ \hline & 01 \end{array}$$

37 (d) $5C8_{16} = 0101\ 1100\ 1000_2$

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40 (f) $2890_{10} = ?_{16}$

Repeated Division by 16

$2890 \div 16 = 180$ $\quad \boxed{10}$

$180 \div 16 = 11$ $\quad \boxed{4}$

$11 \div 16 = 0$ $\quad \boxed{11}$

$B\ 4\ A_{16}$

---

42 (b)

$$\begin{array}{r} \overset{1}{C}\ \cancel{8}^{18_{16} = 24_{10}} \\ -\ 3A = 10_{10} \\ \hline 8E \quad 14_{10} \end{array}$$

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43 (g) $163_8 = ?_{10}$

$= 1 * 8^2 + 6 * 8^1 + 3 * 8^0 = 115_{10}$

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46 (h) $\underbrace{0}\underbrace{1011}\underbrace{0000}\underbrace{011}_2 = ?_8$

$2\ 6\ 0\ 3_8$